

Chapter 4 Relations and Digraphs

Preliminaries On Matrices

Definition 4.1

A **matrix** is a rectangular array of numbers, denoted by

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

The numbers in the array are called the **entries** in the matrix.

Note

- (1) The **size** of a matrix is described in terms of the number of rows and columns it contains.
If a matrix A has m rows and n columns, then A is an m by n matrix (written $m \times n$).
- (2) A matrix A with n rows and n columns is called a **square matrix of order n** and the entries $a_{11}, a_{22}, \dots, a_{nn}$ are said to be on the **main diagonal** of A .
- (3) The entry in row i and column j of a matrix A is commonly denoted by a_{ij} and we write $A = [a_{ij}]$.

Matrix Operations

Definition 4.2

Let A and B be matrices of the same size.

The **sum** $A + B$ is the matrix obtained by adding the entries of B to the corresponding entries of A .

The **difference** $A - B$ is the matrix obtained by subtracting the entries of B from the corresponding entries of A .

Definition 4.3

If A is an $m \times r$ matrix and B is an $r \times n$ matrix, then the **product** AB is the $m \times n$ matrix whose entries are determined as follows:

To find the entry in row i and column j of AB , single out row i from the matrix A and column j from the matrix B . Multiply the corresponding entries from the row and column together and then add up the resulting products.

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$$AB = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1r} \\ a_{21} & a_{22} & \cdots & a_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ir} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mr} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{r1} & b_{r2} & \cdots & b_{rj} & \cdots & b_{rn} \end{bmatrix} = \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & c_{ij} & \vdots \\ c_{m1} & \cdots & c_{mn} \end{bmatrix}$$

where $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \dots + a_{ir}b_{rj} = \sum_{k=1}^r a_{ik}b_{kj}$

Example 4.1

$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} -2 & 5 \\ 4 & -3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 6 & 16 \end{bmatrix}$$

Note

AB is defined only when the number of columns of A is the same as the number of rows of B .

Transpose of a Matrix

Definition 4.4

If A is any $m \times n$ matrix, then the transpose of A , denoted by A^T , is defined to be the $n \times m$ matrix that results from interchanging the rows and columns of A .

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}.$$

Note

A square matrix A is called **symmetric** if $A^T = A$.

Boolean Matrix Operations

Definition 4.5

A **Boolean matrix** is an $m \times n$ matrix whose entries are either “0” or “1”.

Join, Meet and Boolean Product

Definition 4.6

(1) Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ Boolean matrices.

(a) The **join** of A and B , $A \vee B = C = [c_{ij}]$ where

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0 & \text{if } a_{ij} \text{ and } b_{ij} \text{ are both } 0 \end{cases}$$

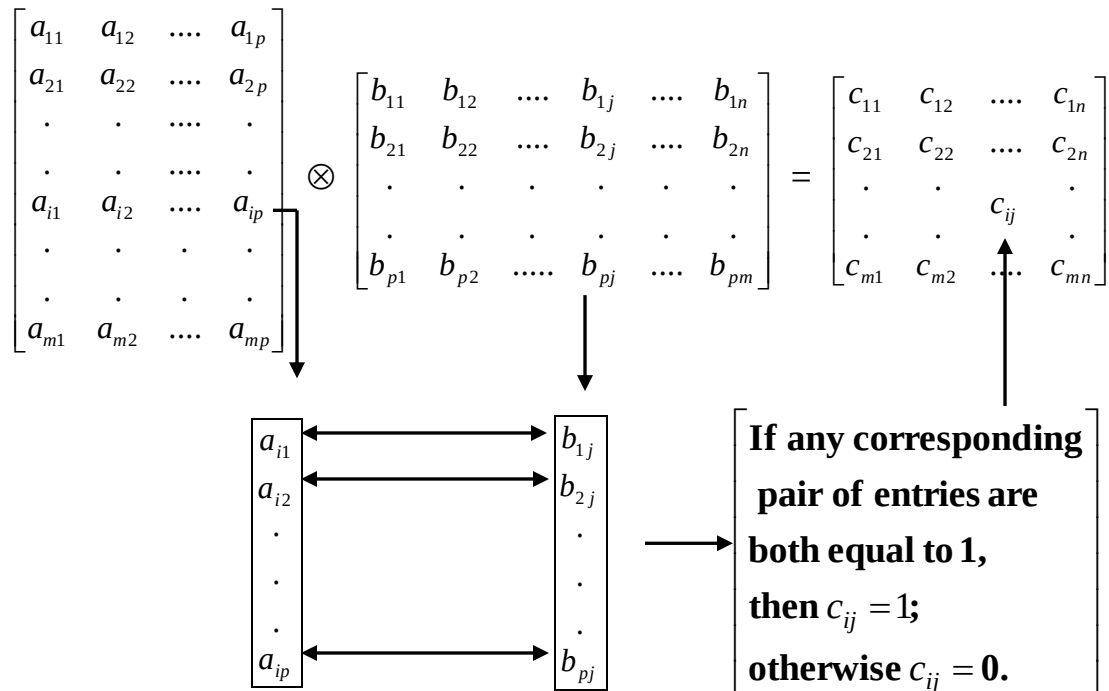
(b) The **meet** of A and B , $A \wedge B = D = [d_{ij}]$ where

$$d_{ij} = \begin{cases} 1 & \text{if } a_{ij} \text{ and } b_{ij} \text{ are both } 1 \\ 0 & \text{if } a_{ij} = 0 \text{ or } b_{ij} = 0 \end{cases}$$

(2) Suppose that $A = [a_{ij}]$ is an $m \times p$ Boolean matrix and $B = [b_{ij}]$ is a $p \times n$ Boolean matrix.

The **Boolean product** of A and B , denoted $A \otimes B$, is the $m \times n$ Boolean matrix $C = [c_{ij}]$ defined by

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 \text{ and } b_{kj} = 1 \text{ for some } k, 1 \leq k \leq p \\ 0 & \text{otherwise} \end{cases}$$



Example 4.2

$$\text{Let } A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ and } C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix}.$$

Compute the following:

- (a) $A \vee B$
- (b) $B \wedge A$
- (c) $A \vee (B \wedge A)$
- (d) $A \otimes C$

$$(a) \quad A \vee B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \vee \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} =$$

$$(b) \quad B \wedge A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \wedge \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} =$$

$$(c) \quad A \vee (B \wedge A) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \vee \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} =$$

$$(d) \quad \text{Let } [d_{ij}] = A \otimes C$$

$$d_{11} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \quad d_{12} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} =$$

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$$d_{13} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} =$$

$$d_{14} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} =$$

$$d_{21} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} =$$

$$d_{22} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} =$$

$$d_{23} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} =$$

$$d_{24} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} =$$

$$d_{31} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} =$$

$$d_{32} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} =$$

$$d_{33} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} =$$

$$d_{34} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} =$$

$$d_{41} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} =$$

$$d_{42} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} =$$

$$d_{43} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} =$$

$$d_{44} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} =$$

So $A \otimes C =$

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Preliminaries On Set Theory

Definition 4.7

A **set** is any well-defined list, collection, or class of objects.

The objects comprising the set are called its **elements** or **members**.

The statement “ p is an element of A ” is written $p \in A$.

Subsets

Definition 4.8

If A and B are sets, A is called a subset of B , written $A \subseteq B$, if and only if every element of A is also an element of B .

More specifically, A is a subset of B if $x \in A$ implies $x \in B$.

Power Set

Definition 4.9

The family of all the subsets of any set A is called the **power set** of A .

Notation: $P(A)$.

Example 4.3

(a) Let $A = \{a, b\}$.

Then $P(A) =$

(b) Let $B = \{4, 7, 8\}$.

Then $P(B) =$

Product Sets

Definition 4.10

For any nonempty sets A and B , the **product set** or **Cartesian product** of A and B is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Note

- 1) For any two finite, nonempty sets A and B , $|A \times B| = |A| |B|$
($|A|$ represents the cardinality of A or the number of elements in A)
- 2) In general, $A \times B \neq B \times A$
- 3) For n nonempty sets $A_1, A_2, \dots, A_n$, the **product set** or **Cartesian product** is

$$A_1 \times A_2 \times \dots \times A_n = \{(x_1, x_2, \dots, x_n) \mid x_i \in A_i, i = 1, 2, \dots, n\}$$

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Example 4.4

A marketing research firm classifies a person according to the following two criteria:

Gender: male (m); female (f).

Highest level of education completed: secondary school (s); high school (h); college (c); university (u).

Let $G = \{m, f\}$ and $E = \{s, h, c, u\}$

- (i) Find $G \times E$ and $E \times G$.
- (ii) Interpret the classification (m, h)

Example 4.5

Let $A = \{a, b, c\}$, $B = \{1, 2\}$ and $C = \{\#, *\}$. List all the elements of $A \times B \times C$.

Partitions

Definition 4.11

Let $A_1, A_2, \dots, A_n$ be the subsets of a nonempty set A , a **partition** or **quotient set** of a set A is denoted by

$$\mathcal{P}(A) = \{A_1, A_2, \dots, A_n\} \text{ if and only if}$$

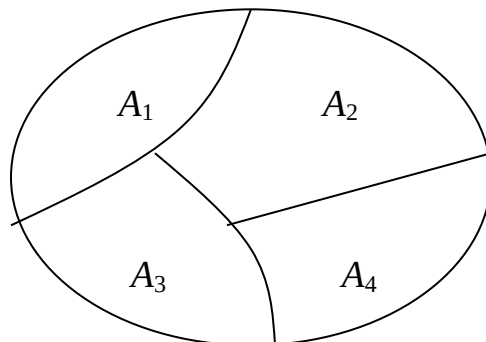
(i) $A_1 \cup A_2 \cup \dots \cup A_n = A$

(ii) $A_i \cap A_j = \emptyset$ for $A_i \neq A_j$

Each A_i is called a **block** or **cell** of $\mathcal{P}$.

Example 4.6

(1)



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- (2) Let $A = \{1, 2, 3\}$
- (a) List all partitions of A with 2 blocks.
 - (b) List all partitions of A .
- (3) Let A be a set of all positive integers and B be the set of all negative integers. Is $\{A, B\}$ a partition of $\mathbb{Z}$?
- (4) Let $A = \{a, b, c, d, e, f, g\}$. Which of the following is/are partition(s) of A ? Give reason(s) to support your answer.
- (i) $\{\{a, b, e\}, \{c, f\}, \{d, g\}\}$
 - (ii) $\{\{a, b\}, \{c, d, g\}, \{d, e, f\}\}$
 - (iii) $\{\{a, c, f\}, \{d, e\}\}$
- (5) List all partition of $B = \{1, 2, 3, 4\}$.
- $\{\{1, 2, 3, 4\},$
 $\{\{1\}, \{2, 3, 4\}\}, \{\{2\}, \{1, 3, 4\}\}, \{\{3\}, \{1, 2, 4\}\}, \{\{4\}, \{1, 2, 3\}\},$
 $\{\{1\}, \{2\}, \{3, 4\}\}, \{\{1\}, \{3\}, \{2, 4\}\}, \{\{1\}, \{4\}, \{2, 3\}\},$
 $\{\{2\}, \{3\}, \{1, 4\}\}, \{\{2\}, \{4\}, \{1, 3\}\}, \{\{3\}, \{4\}, \{1, 2\}\},$
 $\{\{1\}, \{2\}, \{3\}, \{4\}\},$
 $\{\{1, 2\}, \{3, 4\}\}, \{\{1, 3\}, \{2, 4\}\}, \{\{1, 4\}, \{2, 3\}\},$

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Note

The number of partitions of a set with n elements into k subsets satisfies the recurrence relation $S(n, k) = S(n-1, k-1) + k \cdot S(n-1, k)$ with initial conditions $S(n, 1) = S(n, n) = 1$.

Example 4.7

- (a) Let $A = \{1, 2, 3\}$ in Example 3.3(2), so $n = 3$
 $S(3, 1) = S(3, 3) = 1$
 $S(3, 2) = S(2, 1) + 2S(2, 2) = 1 + 2(1) = 3$
So all together we have $1 + 1 + 3 = 5$ partition
- (b) Let $A = \{1, 2, 3, 4\}$ in Example 3.3(5), so $n = 4$.
 $S(4, 1) = S(4, 4) = 1$
 $S(4, 3) = S(3, 2) + 3S(3, 3) = 3 + 3(1) = 6$
 $S(4, 2) = S(3, 1) + 2S(3, 2) = 1 + 2(3) = 7$
So all together we have $1 + 1 + 6 + 7 = 15$ partition

Relations

The notation of a relation between two sets of objects is quit common and important. For human being, throughout life, we experience a wide variety of personal relationships. The words such as “father of”, “son of”, “husband of”, “friend of”, etc are about relationships between two persons. For two objects, typical comparisons using such words as “bigger than”, “better than”, “faster than”, etc. In this section, we learn that elements within a set or different sets often have a special connection with one another that can be described as a relation. Here, we shall focus our attention to **binary relation**, a relation between two elements.

Definition 4.12

Let A and B be two nonempty sets and let $a \in A, b \in B$.

A **binary relation R from A to B** is a subset of $A \times B$ ($R \subseteq A \times B$)

A **binary relation R on A** is a subset of $A \times A$ ($R \subseteq A \times A$)

Given an ordered pair (a, b) in $A \times B$, **a is related to b by R** , written $a R b$, if and only if, (a, b) is in R .

Notation

$$\begin{aligned} a R b &\Leftrightarrow (a, b) \in R && (a \text{ is related to } b \text{ by } R) \\ a \not R b &\Leftrightarrow (a, b) \notin R && (a \text{ is not related to } b \text{ by } R) \end{aligned}$$

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Example 4.8

Let $A = \{\text{Kuala Lumpur } (k), \text{ Jakarta } (j), \text{ Petaling Jaya } (p), \text{ Shanghai } (s), \text{ London } (l)\}$

$B = \{\text{Malaysia } (m), \text{ Thailand } (t), \text{ China } (c), \text{ Indonesia } (i)\}$

- (i) Let R_1 be the relation from A to B consisting of ordered pairs (a, b) such that a is the capital of b . List the elements of R_1 .

- (ii) Let R_2 be the relation from A to B consisting of ordered pairs (x, y) such that x is a city of y . List the elements of R_1 .

Example 4.9

Let $A = \{3, 4, 5, 6, 7\}$ and let R be the relation (less than) on A . List the elements of R .

Example 4.10

Let R be a relation on A defined by: $x R y \Leftrightarrow x$ divides y .

- (a) If $A = \{1, 2, 3, 4\}$, list all elements in R .

- (b) If $A = \mathbb{Z}$, determine whether the following is true or false.
 - (i) $4 R 12$
 - (ii) $3 R 7$
 - (iii) $5 R 0$
 - (iv) $12 R 4$
 - (v) $63 R 3$
 - (vi) $0 R 0$

Example 4.11

Let $A = \{0, 2, 3, 4, 10, 12\}$ and $B = \{0, 1, 3, 4\}$. Let R be the relation from A to B defined by: $x R y \Leftrightarrow x$ is even.

Find R .

Domain and Range of R

Definition 4.13

Let $R \subseteq A \times B$ be a relation from A to B .

The **domain of R** : $\text{Dom}(R) = \{a \in A \mid a R b \text{ for some } b \in B\}$
= The set of all the a values of (a, b) in R .

The **range of R** : $\text{Ran}(R) = \{b \in B \mid a R b \text{ for some } a \in A\}$
= The set of all the b values of (a, b) in R .

Example 4.12

Find the domain and range of the following relations R :

- (a) $R = \{(a, a), (a, b), (b, b), (b, c), (d, c), (d, e)\}$
- (b) $A = \{-2, -1, 0, 1, 2, 3\}$, $B = \{0, 1, 4\}$ and define the relation R from A to B as $x R y \Leftrightarrow y = x^2$.

Definition 4.14

Let $R \subseteq A \times B$ be a relation from A to B and $a \in A$.

The **R -relative set of a** is $R(a) = \{b \in B \mid a R b\}$

Example 4.13

- (1) Let R be the relation in Example 4.12(a). Find $R(a)$, $R(d)$ and $R(\{a, d\})$.
- (2) Let R be the relation in Example 4.12(b). Find $R(-2)$, $R(3)$ and $R(\{-1, 0, 2\})$.

The Matrix of a Relation

Definition 4.15

Let $A = \{a_1, a_2, \dots, a_m\}$ and $B = \{b_1, b_2, \dots, b_n\}$ and let R be a relation from A to B . We represent R by the $m \times n$ matrix $\mathbf{M}_R = [m_{ij}]$, which is defined by

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

The matrix $\mathbf{M}_R$ is a Boolean matrix and is called the **matrix of R** .

Example 4.14

- (a) Let $A = \{a, b, c, d, e, f\}$, $B = \{1, 2, 4, 5, 6\}$ and let $R = \{(a, 2), (a, 5), (b, 5), (b, 6), (c, 1), (c, 4), (e, 5), (f, 6)\}$ be the relation from A to B . Find the matrix of R .

- (b) Let $M_R = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$ be the matrix of R and R is a relation from $A = \{a, b, c\}$ to $B = \{1, 2, 3, 4\}$. Find the relation R .

The Diagraph of a Relation

Definition 4.16

If R is a relation on a finite set A , we can also represent R pictorially as follows:

- (i) Draw a small circle for each element of A and label the circle with the corresponding element of A . Such a circle is called a **vertex** (**vertices** for circles).
- (ii) Draw a line with an arrow from vertex v_i to vertex v_j if and only if $v_i R v_j$. This line (v_i, v_j) is called an edge. An edge of the form (x, x) is called a **loop**.
- (iii) The resulting pictorial representation of R is called a **directed graph** or **digraph**.

Example 4.15

Let $A = \{1, 2, 4, 5, 6\}$ and a relation R on A be defined by $x R y \Leftrightarrow x + y > 8$. Draw the digraph for R and find the matrix of R .

Example 4.16

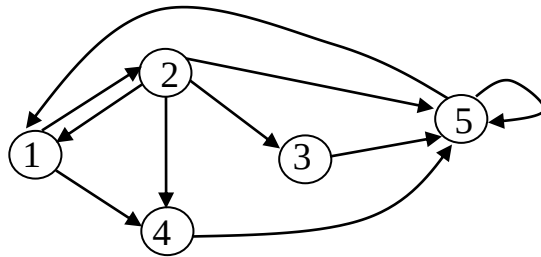
Let $A = \{a, b, c, d, e\}$ and let R be the relation on A that has the matrix below:

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

Draw the digraph for R .

Example 4.17

Find the relation determined by the following digraph:



Definition 4.17

If R is a relation on a finite set A and $a, b \in A$, then

- (i) the **in-degree of a** is the number of b such that $(b, a) \in R$.
- (ii) the **out-degree of a** is the number of b such that $(a, b) \in R$.

Note

- 1) in-degree ----- the number of arrows pointing inwards to a or the number of “1” in the column corresponds to a of M_R .
- 2) out-degree ----- the number of arrows pointing outwards from a or the number of “1” in the row corresponding to a of M_R

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Example 4.18

Find the in-degree and out-degree of each vertex of the digraphs in Example 4.16 and 4.17.

Example 4.16

Vertex	a	b	c	d	e
In-degree					
Out-degree					

Example 4.17

Vertex	1	2	3	4	5
In-degree					
Out-degree					

Restriction of R to B

Definition 4.18

If R is a relation on a finite set A and $B \subseteq A$, the **restriction of R to B** is

$$R \cap (B \times B).$$

(keep the relations of the points in B only)

Example 4.19

Let $A = \{1, 2, 3, 5, 6\}$, $B = \{2, 3\}$ and a relation R on A be defined by

$$x R y \leftrightarrow xy \leq 12.$$

Find the restriction of R to B .

Paths in relations and digraphs

Definition 4.19

Suppose that R is a relation on a set A .

A **path** from a vertex a to a vertex b is a sequence of vertices

$$a, v_1, v_2, \dots, v_{n-1}, b$$

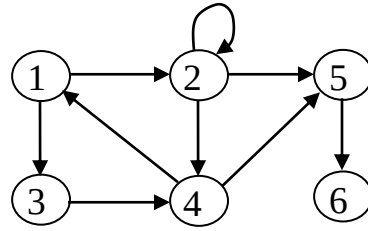
such that $a R v_1, v_1 R v_2, v_2 R v_3, \dots, v_{n-1} R b$.

The **length of a path** is the number of edges in that path.

A **cycle** is a path begins and ends at the same vertex. ($a = b$)

Example 4.20

Consider the following digraph.



- (i) Find path(s) from the vertex u to the vertex v with given length.
 - (a) $u = 1, v = 4$, length = 2;
 - (b) $u = 2, v = 6$, length = 3.
 - (c) $u = 5, v = 3$, length = 2.
- (ii) Find all cycles with length 3.

Definition 4.20

Let n be a fixed positive integer.

- (1) A **relation R^n** on a set A is defined as follows:
 $x R^n y$ means that there exists a path of length n from x to y .

- (2) A **relation R^∞** on a set A is defined as follows:
 $x R^\infty y$ means that there is some path from x to y .
 $R^\infty = R^1 \cup R^2 \cup R^3 \cup \dots$

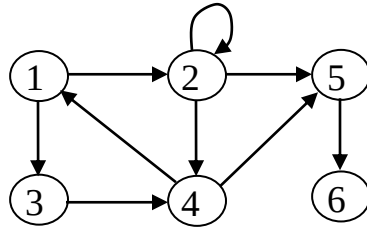
The relation R^∞ is called the **connectivity relation** for R .

- (3) A **relation R^*** of R on a set A is defined as follows:
 $x R^* y$ means that $x = y$ or $x R^\infty y$.
 The relation R^* is called the **reachability relation** of R .
 In General, if R is a relation on a set $A = \{x_1, x_2, \dots, x_n\}$, then

$$R^* = R^\infty \cup \{(x_1, x_1), (x_2, x_2), \dots, (x_n, x_n)\}$$

Example 4.21

Consider the following digraph. Compute R^2 .



Example 4.22

Let $A = \{a, b, c, d, e, f\}$ and $R = \{(a, a), (a, b), (b, c), (c, d), (c, e), (d, e)\}$. Compute R^2 , R^∞ and R^* .

Theorem 4.1

Let $n \geq 2$ and R be a relation on a finite set A , we have

$$M_{R^n} = M_R \otimes M_R \otimes \cdots M_R = (M_R)_{\otimes}^n \quad (n \text{ factors})$$

$$M_{R^\infty} = M_R \vee M_{R^2} \vee M_{R^3} \vee \cdots$$

$$M_{R^*} = M_{R^\infty} \vee I_n, \text{ where } I_n \text{ is the } n \times n \text{ identity matrix.}$$

Example 4.23

Let $R = \{(a, a), (a, c), (b, a), (b, d), (a, b), (d, c)\}$.

Compute M_{R^2} , M_{R^3} , M_{R^4} and M_{R^∞} .

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{R^2} &= M_R \otimes M_R \\ &= \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} M_{R^3} &= M_R \otimes M_{R^2} \\ &= \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
 M_{R^4} &= M_R \otimes M_{R^3} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = M_{R^3}
 \end{aligned}$$

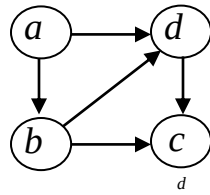
Since $M_{R^4} = M_{R^3}$, we see that $R^3 = R^4$ and $R^3 = R^4 = R^5 = R^6 = \dots$

$$\therefore R^\infty = R^1 \cup R^2 \cup R^3 \cup R^4 \cup R^5 \cup \dots = R^1 \cup R^2 \cup R^3$$

$$\begin{aligned}
 M_{R^\infty} &= M_{R^1} \vee M_{R^2} \vee M_{R^3} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

Example 4.24

Let $A = \{a, b, c, d\}$ and let R be the relation whose digraph is shown as below. Compute M_{R^∞} , M_{R^*} and hence find R^∞ and R^* .



Properties of Relations

Reflexive and irreflexive

Definition 4.21

- (1) A relation R on a set A is **reflexive** if $(a, a) \in R \forall a \in A$.
(every element $a \in A$ is related to itself)
- (2) A relation R on a set A is **irreflexive** if $(a, a) \notin R \forall a \in A$.
(no element in A is related to itself)

Note

- (1) If R on A is reflexive, then

$$(i) \quad M_R = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{pmatrix} \quad (\text{All 1's on its main diagonal})$$

- (ii) the digraph of R has a cycle of length 1 at every vertex.

- (2) If R on A is irreflexive, then

$$(i) \quad M_R = \begin{pmatrix} 0 & & & \\ & 0 & & \\ & & 0 & \\ & & & \ddots \\ & & & & 0 \end{pmatrix} \quad (\text{All 0's on its main diagonal})$$

- (ii) the digraph of R has no cycle of length 1.

- (3) Not irreflexive **does not mean** reflexive.

- (4) If R on A is not reflexive and not irreflexive, then there exist some 0's and some 1's on the main diagonal of M_R .

Example 4.25

Let $A = \{a, b, c\}$ and $R = \{(a, a), (a, c), (b, b)\}$.

R is not reflexive since $(c, c) \notin R$.

R is not irreflexive since $(a, a) \in R$.

$$M_R = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Example 4.26

Let $A = \{1, 2, 3, 4, 5, 6\}$.

Determine whether the following relations are reflexive, irreflexive or not reflexive and not irreflexive. Give reason(s) to support your answer.

(a) $R = \{(1, 1), (1, 2), (2, 2), (3, 3), (4, 4), (5, 5), (5, 6), (6, 6)\}$.

(b) $R = \{(1, 1), (2, 2), (3, 4), (4, 4), (5, 5), (6, 6)\}$.

(c) $R = \{(x, y) \in A \times A \mid x > y\}$.

Symmetric, Asymmetric and Antisymmetric

Definition 4.22

- (1) A relation R on a set A is **symmetric** if $(a, b) \in R$ then $(b, a) \in R$.
 $(\forall (a, b) \in R, a R b \Rightarrow b R a)$
- (2) A relation R on a set A is **asymmetric** if $(a, b) \in R$ then $(b, a) \notin R$.
 $(\forall (a, b) \in R, a R b \Rightarrow b \not R a)$
- (3) A relation R on a set A is **antisymmetric** if $(a, b) \in R$ and $(b, a) \in R$ then $a = b$.
 $(\forall (a, b) \in R, a \neq b \Rightarrow a \not R b \text{ or } b \not R a)$

Note

- (1) R is **not** symmetric if we have **some** a and $b \in A$ with $a R b$, but $b \not R a$.
- (2) R is **not** asymmetric if we have **some** a and $b \in A$ with both $a R b$ and $b R a$.
- (3) R is **not** antisymmetric if we have **some** a and $b \in A$, $a \neq b$, and both $a R b$ and $b R a$.

Note

- (1) If R on A is symmetric, then
 - (i) $M_R = M_R^T$
 - (ii) the digraph of R has the property that if there is an edge from vertex i to vertex j , then there is an edge from vertex j to vertex i . Thus all edges are “two-way”.
- (2) If R on A is asymmetric, then
 - (i) $M_R = [m_{ij}]$ satisfies the property that
$$m_{ii} = 0 \text{ for all } i, \text{ and}$$
$$\text{if } m_{ij} = 1 \text{ then } m_{ji} = 0$$
 - (ii) the digraph of R cannot simultaneously have an edge from vertex i to vertex j and an edge from vertex j to vertex i . Thus there can be no cycles of length 1, and all edges are “one-way”.
- (3) If R on A is antisymmetric, then
 - (i) $M_R = [m_{ij}]$ satisfies the property that if $i \neq j$, then $m_{ij} = 0$ or $m_{ji} = 0$.
 - (ii) the digraph of R has the property that for different vertices i and j there cannot be an edge from vertex i to vertex j and an edge from vertex j to vertex i . When $i = j$, no condition is imposed. Thus there may be cycles of length 1, but again all edges are “one way”.
- (4) Not symmetric **does not mean** asymmetric or antisymmetric.
(The terms symmetric and antisymmetric are not opposites)

Example 4.27

Let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (2, 1), (1, 3)\}$.

Example 4.28

Let $A = \{1, 2, 3, 4, 5, 6\}$.

Determine whether the following relations are symmetric, asymmetric, antisymmetric or none of them. Give reason(s) to support your answer.

(a) $R = \{(1, 2), (2, 1), (2, 4), (3, 4), (4, 2), (4, 3), (5, 6), (6, 5), (6, 6)\}$.

(b) $R = \{(1, 2), (2, 1), (2, 3), (3, 2), (3, 4), (5, 5)\}$.

(c) $R = \{(a, b) \in A \times A \mid a > b\}$.

Graphs

Definition 4.23

Let u and v be vertices of a digraph.

If u and v must always be connected by edges in both directions (if there is an edge from u to v , then there is an edge from v to u), then these two edges will be replaced by an undirected edge (an edge without arrow).

The resulting diagram will be called a **graph** (or **undirected graph**).

Note

The symmetric relation can be represented by a graph

Example 4.29

Draw the digraph and graph of the symmetric relation R in Example 4.28(a).

$R = \{(1, 2), (2, 1), (2, 4), (3, 4), (4, 2), (4, 3), (5, 6), (6, 5), (6, 6)\}$.

Transitive

Definition 4.24

A relation on R on a set A is **transitive** if

$$(a, b) \in R \text{ and } (b, c) \in R, \text{ then } (a, c) \in R.$$

$$(\forall a, b, c \in A, a R b \text{ and } b R c \Rightarrow a R c)$$

Note

- (1) A relation R on a set A is **not transitive** if there exist $a, b, c \in A$ so that $(a, b) \in R$ and $(b, c) \in R$ but $(a, c) \notin R$.

If such a, b and c do not exist, then R is transitive.

- (2) If R on A is transitive, then

- (i) $M_R = [m_{ij}]$ has the property
if $m_{ij} = 1$ and $m_{jk} = 1$ then $m_{ik} = 1$

Thus the transitive of R means that if $(M_R)_{\otimes}^2$ has a 1 in any position, then M_R must have 1 in the same position.

The converse is not true.

- (ii) the diagraph of R has the property that if there is a path of length two from a vertex u to a vertex v , then there is an edge from u to v .

$$(u, v) \in R^2 \Rightarrow (u, v) \in R$$

In other words, if $u R^2 v$ then $u R v$. (The converse is not necessary true)

Example 4.30

Let $A = \{1, 2, 3, 4, 5, 6\}$. Determine whether the following relations are transitive or not. Give reason(s) to support your answer.

(a) $R = \{(1, 1), (1, 2), (2, 1), (3, 4), (3, 5), (4, 5)\}.$

(b) $R = \{(1, 2), (2, 1), (2, 2), (2, 3), (3, 2)\}.$

(c) $R = \{(1, 2), (3, 4), (5, 6)\}.$

(d) $R = \{(1, 2), (1, 3), (1, 4), (4, 4)\}$

Example 4.31

Let $A = \{1, 2, 3, 4\}$.

Determine whether each of the following relation is transitive or not.

Discuss your answer by using matrices of relation.

(a) $R = \{(1, 2), (1, 3), (2, 1), (2, 4), (3, 2), (4, 1), (4, 2)\}$

(b) $R = \{(1, 2), (1, 3), (1, 4), (4, 4)\}$.

Example 4.32

	Ref	Irref	Sym	Asym	Antisym	Tran.
$M_{R_1} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$						
$M_{R_2} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$						
$M_{R_3} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$						
$M_{R_4} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$						
$M_{R_5} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$						
$M_{R_6} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$						

Equivalence Relations

Consider the fractions $\frac{1}{3}, \frac{3}{9}, \frac{-5}{-15}, \frac{100}{300}, \dots$ and so on.

They are different ways to represent the same number. They may look different; they may be called different names; but they are all equal. Here, we say the fraction has many equivalent forms. The idea of grouping together things that “look different but are really the same” is the central idea of equivalence relation.

Definition 4.25

A relation R on a set A is called an **equivalence relation** if it is reflexive, symmetric and transitive.

Example 4.33

- (a) Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$. Is R an equivalence relation on A ?
- (b) Is the **inequality relation** “ $<$ ” an equivalence relation?
(Inequality relation “ $<$ ” means $\{(x, y) \in A \times A \mid x < y\}$)

Example 4.34

Let $A = \mathbb{Z}$ and let R be the relation on A defined by $a R b \Leftrightarrow a + b$ is even. Show that R is an equivalence relation.

Divisibility

Definition 4.26

If m and n are integers and $n \neq 0$, then
 m is **divisible by** n if, and only if, $m = qn$ for some integer q .

Notation

$n|m$ ----- n divides m .

Congruent modulo n

Definition 4.27

Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$.

We say **a and b are congruent modulo n** provided that $n|(a-b)$.

We write **$a \equiv b(\text{mod } n)$** which means **$a - b = kn$ for some integer k** .

Example 4.35

- (a) $12 \equiv 7(\text{mod } 5)$?
- (b) $-6 \equiv -8(\text{mod } 4)$?

- (c) $0 \equiv -6(\text{mod } 3)$?

Example 4.36

Let $A = \mathbb{Z}$ and let R be the relation on A defined by

$$R = \{(a, b) \in A \times A \mid a \equiv b(\text{mod } n)\}.$$

Show that R is an equivalence relation.

$$a \equiv b(\text{mod } n) \Leftrightarrow n \mid (a - b) \Leftrightarrow (a - b) = kn \text{ for some integer } k.$$

- (i) $n \mid (a - a) \forall a \in A$.
Hence, $a \equiv a(\text{mod } n)$.
Therefore $(a, a) \in R \forall a \in A$ and R is reflexive.

- (ii) $(a, b) \in R. \Rightarrow a \equiv b(\text{mod } n).$
 $\Rightarrow n \mid (a - b)$
 $\Rightarrow (a - b) = kn$ for some integer k
 $\Rightarrow (b - a) = -kn$ for some integer $-k$.
 $\Rightarrow b \equiv a(\text{mod } n).$
 $\Rightarrow (b, a) \in R$

$\therefore R$ is symmetric.

- (iii) Suppose $(a, b) \in R$ and $(b, c) \in R$.
 $a \equiv b(\text{mod } n)$ and $b \equiv c(\text{mod } n).$
 $\Rightarrow n \mid (a - b)$ and $n \mid (b - c)$
 $\Rightarrow (a - b) = k_1n$ and $(b - c) = k_2n$ for some integers k_1 and k_2

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- $\Rightarrow (a - b) + (b - c) = (k_1 + k_2)n$ for some integers $(k_1 + k_2)$
- $\Rightarrow a - c \equiv (k_1 + k_2)n.$
- $\Rightarrow a \equiv c \pmod{n}$
- $\Rightarrow (a, c) \in R$
- $\Rightarrow R$ is transitive.

Since R is reflexive, symmetric and transitive, we conclude that R is an equivalence relation.

Equivalence Class and Quotient Set

Definition 4.28

Let R_e be an equivalence relation on a set A and let $a \in A$.

The set of all elements that related to a , is called the **equivalence class of a** .

Symbolically: $[a] = \{x \in A \mid (x, a) \in R_e\}$

The set of all the distinct equivalence classes of R form a partition of A . This partition is called the **quotient set of A** . It is denoted by **A/R** .

Example 4.37

Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$ be an equivalence relation on A . Find the equivalence class of each element in A . Then find the quotient set of A .

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Example 4.38

Let $A = \mathbb{Z}^+$ and R be the relation on A defined by

$$R = \{(x, y) \in A \times A \mid x \equiv y \pmod{3}\}.$$

- (i) Find the equivalence class of each element in A .
- (ii) Find the quotient set of A .
- (iii) Show that the quotient set of A is a partition of A .

- (i) $1 \equiv 1 \pmod{3} \equiv 4 \pmod{3} \equiv 7 \pmod{3} \equiv 10 \pmod{3} \equiv \dots$
 $2 \equiv 2 \pmod{3} \equiv 5 \pmod{3} \equiv 8 \pmod{3} \equiv 11 \pmod{3} \equiv \dots$
 $3 \equiv 3 \pmod{3} \equiv 6 \pmod{3} \equiv 9 \pmod{3} \equiv 12 \pmod{3} \equiv \dots$
 $4 \equiv 1 \pmod{3} \equiv 4 \pmod{3} \equiv 7 \pmod{3} \equiv 10 \pmod{3} \equiv \dots$

Hence

$$[1] = \{1, 4, 7, 10, \dots\} = [4] = [7] = \dots = [3k - 2] \quad k = 1, 2, \dots$$

$$[2] = \{2, 5, 8, 11, \dots\} = [5] = [8] = \dots = [3k - 1] \quad k = 1, 2, \dots$$

$$[3] = \{3, 6, 9, 12, \dots\} = [6] = [9] = \dots = [3k] \quad k = 1, 2, \dots$$

- (ii) So there are only three distinct equivalent classes of R , and hence $A/R = \{[1], [2], [3]\}$

- (iii) $[1] \cup [2] \cup [3] = \mathbb{Z}^+$ and
 $[1] \cap [2] = \emptyset, [1] \cap [3] = \emptyset$ and $[2] \cap [3] = \emptyset$
Thus A/R is a partition of A .

Theorem 4.2

Let R_e be an equivalence relation on a set A and let $a, b, c \in A$.

- (1) $a R_e b \Leftrightarrow [a] = [b]$
 $\Leftrightarrow a$ and b are members of the same block of a partition of A .
- (2) $a, b \in [c] \Rightarrow a R_e b$.

Example 4.39

- (1) Let $A = \{1, 2, 3, 4, 5\}$ and consider the partition $\mathcal{P} = \{\{1, 3\}, \{2, 5\}, \{4\}\}$ of A . Find

- (i) the equivalence relation R by listing all the ordered pairs.

- (ii) all the distinct equivalence classes of R .

- (2) Let $A = \{1, 2, 3, 4\}$ and consider the partition $\mathcal{P} = \{\{1, 2, 3\}, \{4\}\}$ of A . Find the equivalence relation R on A determined by $\mathcal{P}$.

Manipulation of Relation

Let R and S be relations from a set A to a set B . Then R and S are simply subsets of $A \times B$, we can use set operations on R and S as below.

Definition 4.29

Let R and S be relations from a set A to a set B and $x \in A, y \in B$.

- (1) **Intersection relation:** $x (R \cap S) y$ means $x R y \wedge x S y$
- (2) **Union relation:** $x (R \cup S) y$ means $x R y \vee x S y$
- (3) **Complementary relation:** $x \bar{R} y \Leftrightarrow x \not R y$.
- (4) **Inverse relation:** $x R^{-1} y \Leftrightarrow y R x$.

The matrix of the above relations:

- (1) $M_{R \cup S} = M_R \vee M_S$
- (2) $M_{R \cap S} = M_R \wedge M_S$
- (3) $(M_R)_{ij} = 1 \Leftrightarrow (M_{\bar{R}})_{ij} = 0$
- (4) $M_{R^{-1}} = (M_R)^T$

Example 4.40

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$. Let R and S be relations on $A \times B$ defined by

$$R = \{(1, a), (1, b), (2, b), (2, c), (3, b), (4, a)\} \text{ and}$$

$$S = \{(1, b), (2, c), (3, b), (4, b)\}.$$

Compute

- (i) $R \cap S$ (ii) $R \cup S$ (iii) $\bar{R}$ (iv) R^{-1}

(i) $R \cap S = \{(1, b), (2, c), (3, b)\}$

(ii) $R \cup S = \{(1, a), (1, b), (2, b), (2, c), (3, b), (4, a), (4, b)\}$

(iii) $\bar{R} = \{(1, c), (1, d), (2, a), (2, d), (3, a), (3, c), (3, d), (4, b), (4, c), (4, d)\}$

(iv) $R^{-1} = \{(a, 1), (b, 1), (b, 2), (c, 2), (b, 3), (a, 4)\}$

Example 4.41

Let $A = \{1, 2, 3\}$ and let R and S be relations on A .

Suppose $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $M_S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Find $M_{R \cup S}$, $M_{R \cap S}$, $M_{\bar{R}}$

and $M_{R^{-1}}$.

$$M_{R \cup S} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad M_{R \cap S} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$M_{\bar{R}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad M_{R^{-1}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Transitive Closure and Warshall's algorithm

If a relation R on a set A fails to have a certain property such as reflexivity, symmetry and transitivity. We may wish to add some related pairs to R until the new relation R_c has the desired property. If R_c is the smallest such relation (obtained from R by adding the *fewest* related pairs to R) with the desired property, then R_c is called the **closure** of R with respect to that property.

Reflexive closure, symmetry closure and transitive closure.

Example 4.42

Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (1, 2), (1, 3), (2, 3), (3, 1)\}$.

R is not reflexive, not symmetric, and not transitive.

To obtain a reflexive relation from R , we add the pairs $(2, 2)$ and $(3, 3)$ to R .

Since $R_r = \{(1, 1), (1, 2), (1, 3), \boxed{(2, 2)}, (2, 3), (3, 1), \boxed{(3, 3)}\}$ is the smallest reflexive relation contains R , we have R_r is the **reflexive closure** of R .

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To obtain a symmetry relation from R , we add the pairs $(2, 1)$ and $(3, 2)$ to R .

Since $R_s = \{(1, 1), (1, 2), \boxed{(2, 1)}, (2, 3), (1, 3), (3, 1), \boxed{(3, 2)}\}$ is the smallest symmetry relation contains R , we have R_s is the **symmetry closure** of R .

To obtain a transitive relation from R , we add the pairs $(2, 1)$, $(2, 2)$, $(3, 2)$ and $(3, 3)$ to R .

Since $R_t = \{(1, 1), (1, 2), (1, 3), \boxed{(2, 1)}, \boxed{(2, 2)}, (2, 3), (3, 1), \boxed{(3, 2)}, \boxed{(3, 3)}\}$ is the smallest transitive relation contains R , then R_t is the **transitive closure** of R .

To find the transitive closure, we may have a series of steps. Here, we study a theorem and an algorithm to compute the transitive closure of a relation R on a set A .

Theorem 4.3

Let R be a relation on a set A with $|A| = n$.

Then R^∞ is the transitive closure of R and $R^\infty = R^1 \cup R^2 \cup R^3 \cup \dots R^n$.

In other words, powers of R greater than n are not needed to compute R^∞ .

Example 4.43

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 1), (2, 3), (3, 4)\}$.

Find the transitive closure of R .

To find the transitive closure of R , the graphical method is impractical for large sets and relations, and is not systematic.

The matrix method is systematic enough to be programmed for a computer, but it is inefficient and, for large matrices, can be costly.

Here, we study a more efficient algorithm, called **Warshall's algorithm**.

Warshall's algorithm

Let R be a relation on a set A and let $|A| = n$. We know that R^∞ is the transitive closure of R . The purpose of this algorithm is to find R^∞ .

If we define $W_0 = M_R$, then we will have a sequence $W_0, W_1, W_2, \dots, W_n$ where $W_n = M_{R^\infty}$.

We shall learn how to compute each matrix W_k from the previous matrix W_{k-1} . Start with the matrix of R and proceed one step at a time until, in n steps, we reach the matrix of R^∞ . This process is called **Warshall's algorithm**.

Steps:

1. Let $W_0 = M_R$.
2. In general, to compute W_k ,
 - (i) Transfer all 1's in W_{k-1} to W_k .
 - (ii) List the locations $p_1, p_2, p_3, \dots$ in column k of W_{k-1} , where the entry is 1.
 - (iii) List the locations $q_1, q_2, q_3, \dots$ in row k of W_{k-1} , where the entry is 1.
 - (iv) Put 1's in all the positions p_i, q_j of W_k (if they are not already there).
 - (v) Using this recurrence relation, find W_k for $k = 1, 2, 3, \dots, n$. Then $W_n = M_{R^\infty}$. (n is the number of element in A)

Example 4.44

Use Warshall's algorithm to find the transitive closure of R in Example 4.43.

$$W_0 = M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

To compute W_1 , $k = 1$:

Column 1 of W_0 has 1's in location: 2

Row 1 of W_0 has 1's in location: 2

Thus, W_1 is just W_0 with a new 1 in the position (2, 2).

$$\therefore W_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

For W_2 , $k = 2$:

Column 2 of W_1 has 1's in location: 1, 2

Row 2 of W_1 has 1's in location: 1, 2, 3

Thus, W_2 is just W_1 with a new 1 in the position (1, 1) and (1, 3).

$$\therefore W_2 = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

For W_3 , $k = 3$:

Column 3 of W_2 has 1's in location: 1, 2

Row 3 of W_2 has 1's in location: 4

Thus, W_3 is just W_2 with a new 1 in the position (1, 4) and (2, 4).

$$\therefore W_3 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

For W_4 , $k = 4$:

Column 4 of W_3 has 1's in location: 1, 2, 3

Row 4 of W_3 has 1's in location: none

Thus, W_4 is just W_3 with no new 1 is added.

$$\therefore W_4 = W_3$$

$$\text{Since } |A| = 4, M_{R^\infty} = W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The transitive closure of $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4)\}$

Theorem 4.4

If R and S are equivalence relations on a set A , then the smallest equivalence relation contains both R and S is $(R \cup S)^\infty$.

Example 4.45

Let $A = \{1, 2, 3, 4, 5\}$. Given 2 equivalence relations

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4), (5, 5)\}$$

and

$$S = \{(1, 1), (2, 2), (3, 3), (4, 4), (4, 5), (5, 4), (5, 5)\},$$

use Warshall's algorithm to find the smallest equivalence relation contains both R and S .

$$M_R = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, M_S = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

So

$$M_{R \cup S} = M_R \vee M_S = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

To compute $(R \cup S)^\infty$ by Warshall's algorithm, let $W_0 = M_{R \cup S}$.

For $W_1, k = 1$:

Column 1 of W_0 has 1's in location: 1, 2

Row 1 of W_0 has 1's in location: 1, 2

Thus, no new 1's is added to W_1

$$W_1 = W_0$$

For $W_2, k = 2$:

Column 2 of W_1 has 1's in location: 1, 2

Row 2 of W_1 has 1's in location: 1, 2

Thus, no new 1's is added to W_2

$$W_2 = W_1$$

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For W_3 , $k = 3$:

Column 3 of W_2 has 1's in location: 3, 4

Row 3 of W_2 has 1's in location: 3, 4

Thus, no new 1's is added to W_3

$$W_3 = W_2$$

For W_4 , $k = 4$:

Column 4 of W_3 has 1's in location: 3, 4, 5

Row 4 of W_3 has 1's in location: 3, 4, 5

Thus, W_4 is just W_3 with a new 1 in the position (3, 5) and (5, 3)

$$\therefore W_4 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

For W_5 , $k = 5$:

Column 5 of W_4 has 1's in location: 3, 4, 5

Row 5 of W_4 has 1's in location: 3, 4, 5

Thus, no new 1's is added to W_4

$$W_5 = W_4$$

$$\text{Since } |A| = 5, M_{(R \cup S)^\infty} = W_5 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}.$$

Hence

$$(R \cup S)^\infty =$$

Example 4.46

Let $A = \{1, 2, 3, 4, 5\}$ and let R be the relation on A defined by
 $x R y$ if and only if $x + y > 6$.

- (i) List all elements of R .
- (ii) Construct the digraph of R .
- (iii) Determine whether the relation R is reflexive, irreflexive, symmetric, asymmetric, antisymmetric or transitive.